

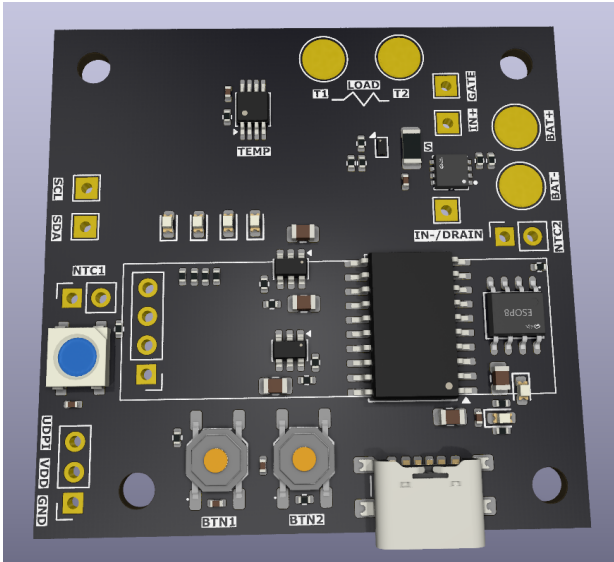
HEATER CONTROLLER

Variant: FIRST PROTOTYPE

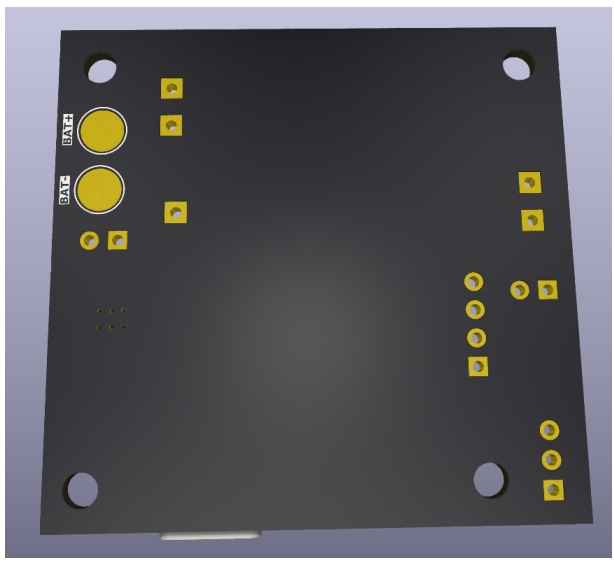
2026-01-22
Rev 1

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TOP VIEW



BOTTOM VIEW



NOTES

- Not fitted components are marked as **X**
- DRAFT - Very early stage of schematic, ignore details.
- PRELIMINARY - Close to final schematic.
- CHECKED - There shouldn't be any mistakes. Contact the engineer if you find any.
- RELEASED - A board with this schematic has been sent to production.

DESIGN CONSIDERATIONS

DESIGN NOTE:

Example text for informational design notes.

DESIGN NOTE:

Example text for debug notes.

DESIGN NOTE:

Example text for cautionary design notes.

DESIGN NOTE:

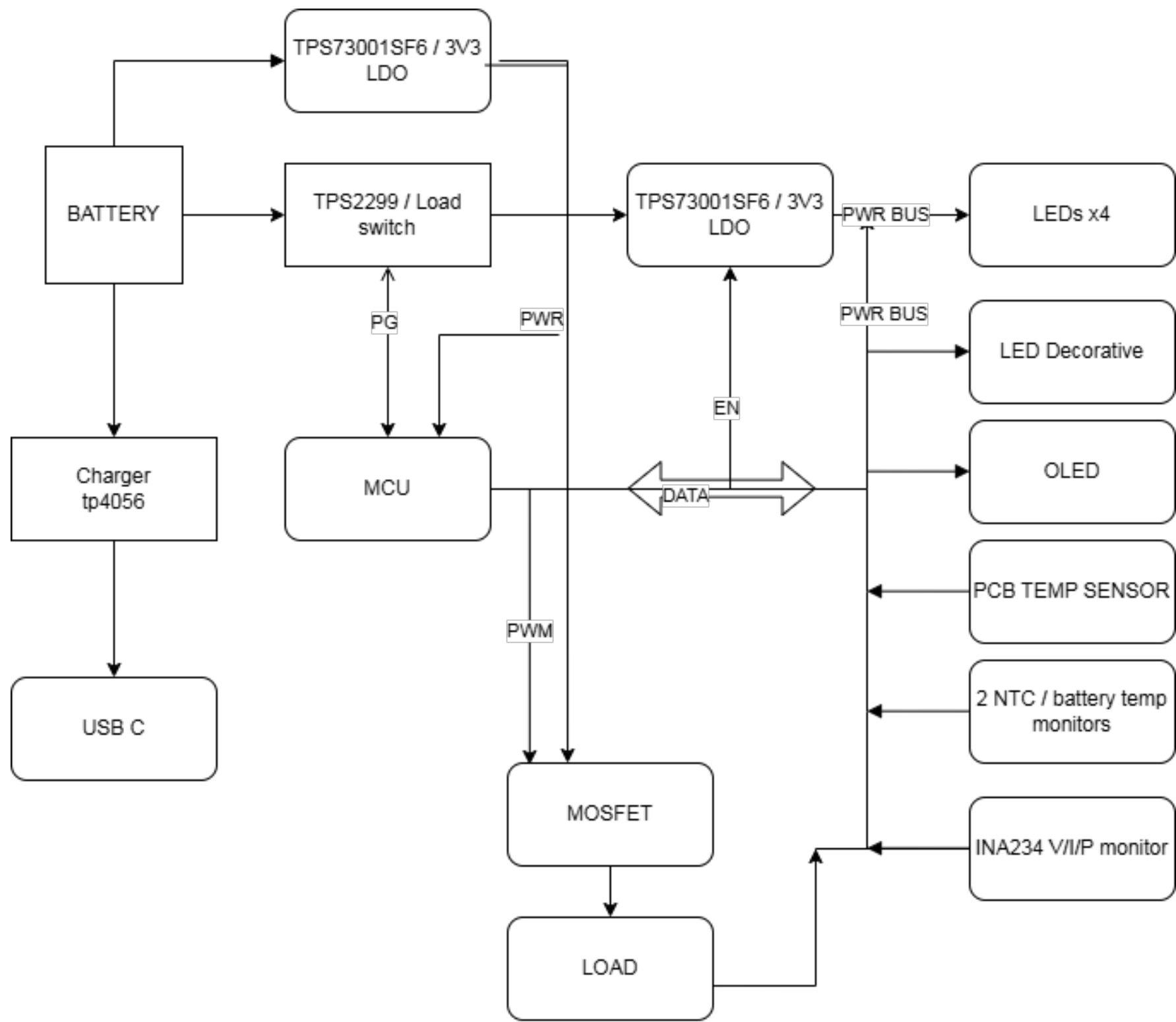
Example text for critical design notes.

LAYOUT NOTE:

Example text for critical layout guidelines.

	Comments:		Company:		Variant:	
			CAPISTOR TECHNOLOGIES - FZCO		FIRST PROTOTYPE	
	Sheet Title:		Board Name:		Project Name:	
	Cover Page		HEATER CONTROLLER		HEATER CONTROLLER	
Sheet Path:		File Name:	Designer:	Date:	Revision:	
/		HEATER CONTROLLER REV_A.kicad	SHOAIB MUSTAFA	Last Modified Date	1	
		Reviewer:		Size:	Sheet:	
		SHOAIB MUSTAFA		A3	1 of 7	

[2] Block Diagram



	Comments:		Company: CAPISTOR TECHNOLOGIES - FZCO		Variant: FIRST PROTOTYPE	
	Sheet Title: Block Diagram		Board Name: HEATER CONTROLLER		Project Name: HEATER CONTROLLER	
	Sheet Path: /Block Diagram/		File Name: Block Diagram.kicad_sch	Designer: SHOAIB MUSTAFA	Date: Last Modified Date	Revision: 1
			Reviewer: SHOAIB MUSTAFA		Size: A3	Sheet: 2 of 7

The image displays two schematic sections from a project architecture document. Section A, titled "Section A - Title A", is shown as a green rectangle labeled "Page 4" and associated with the file "File: Section A - Title A.kicad_sch". It is accompanied by a blue bracket labeled "Description". Section B, titled "Section B - Title B", is shown as a yellow rectangle labeled "Page 5" and associated with the file "File: Section B - Title B.kicad_sch". It is also accompanied by a blue bracket labeled "Description".

	Comments:	Company: CAPISTOR TECHNOLOGIES - FZCO		Variant: FIRST PROTOTYPE	
		Board Name: HEATER CONTROLLER		Project Name: HEATER CONTROLLER	
	Sheet Title: Project Architecture	File Name: Project Architecture.kicad_sch	Designer: SHOAIB MUSTAFA	Date: Last Modified Date	Revision: 1
	Sheet Path: /Project Architecture/		Reviewer: SHOAIB MUSTAFA	Size: A3	Sheet: 3 of 7

- 1- mcu in low power mode 20-60uA
- 2- All gpios to off state
- 3- LDO control turned off
- 4- mosfet gpio off (1uA from mosfet only)
- 5- INA234 only 2.2uA at standby

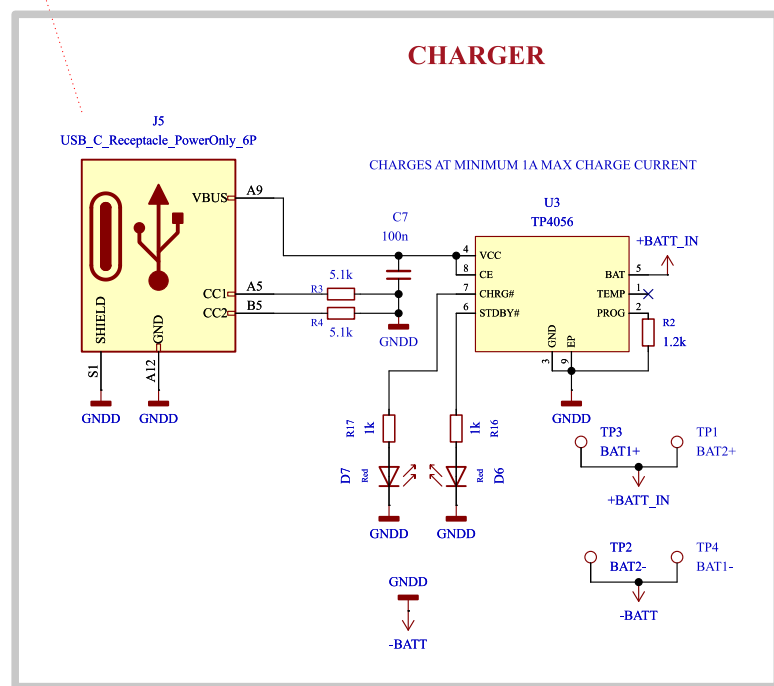
$2.2 + 1 + 60 = 60.3\mu A$

is the max current required for the mcu to stay on and keep the battery current limited acting as UV cutoff

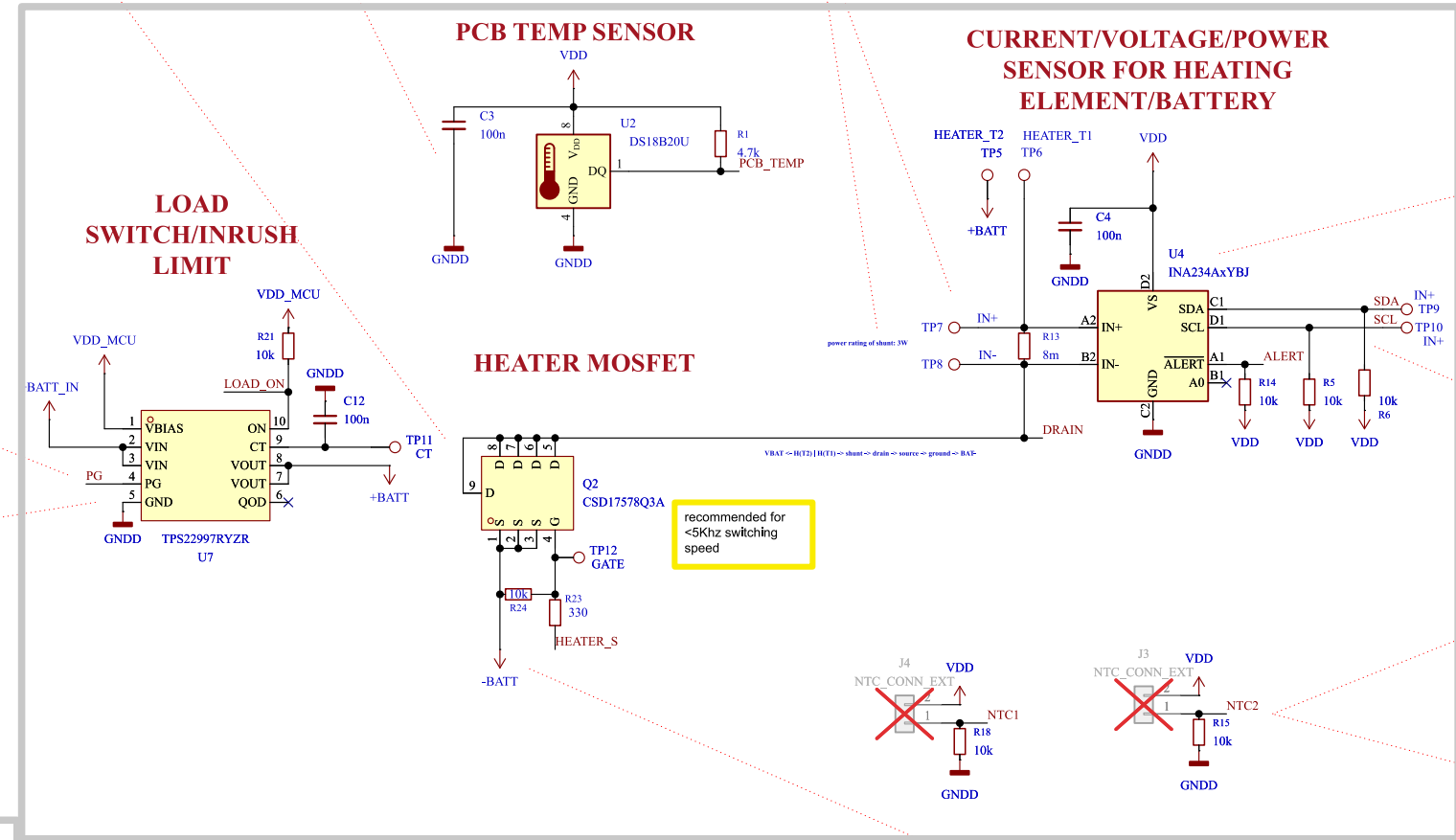
**LAYOUT NOTE:
MUST BE CLOSE TO THE
HEATER PADS TO
MONITOR POINT OF
FAILURE**

SHUNT RESISTOR:
8A max current setting
This value can be changed by changing the shunt resistor
smaller can be selected but will be reduced accuracy

Inrush current capability



- VBias within VDD_MCU range (checked)
- VIN within the VBAT range (checked)
- Rise and fall time configurable around 1.1ms
- 10uA Qs current



monitors Voltage rail getting to mcu. i.e; bat -> ldo -> mcu
Also measures the voltage in the load
Measures the current/power in the load
option to set under, over for supply as well as load bus

Refer to setting up startup register mechanism defined in the datasheet for proper operation in firmware

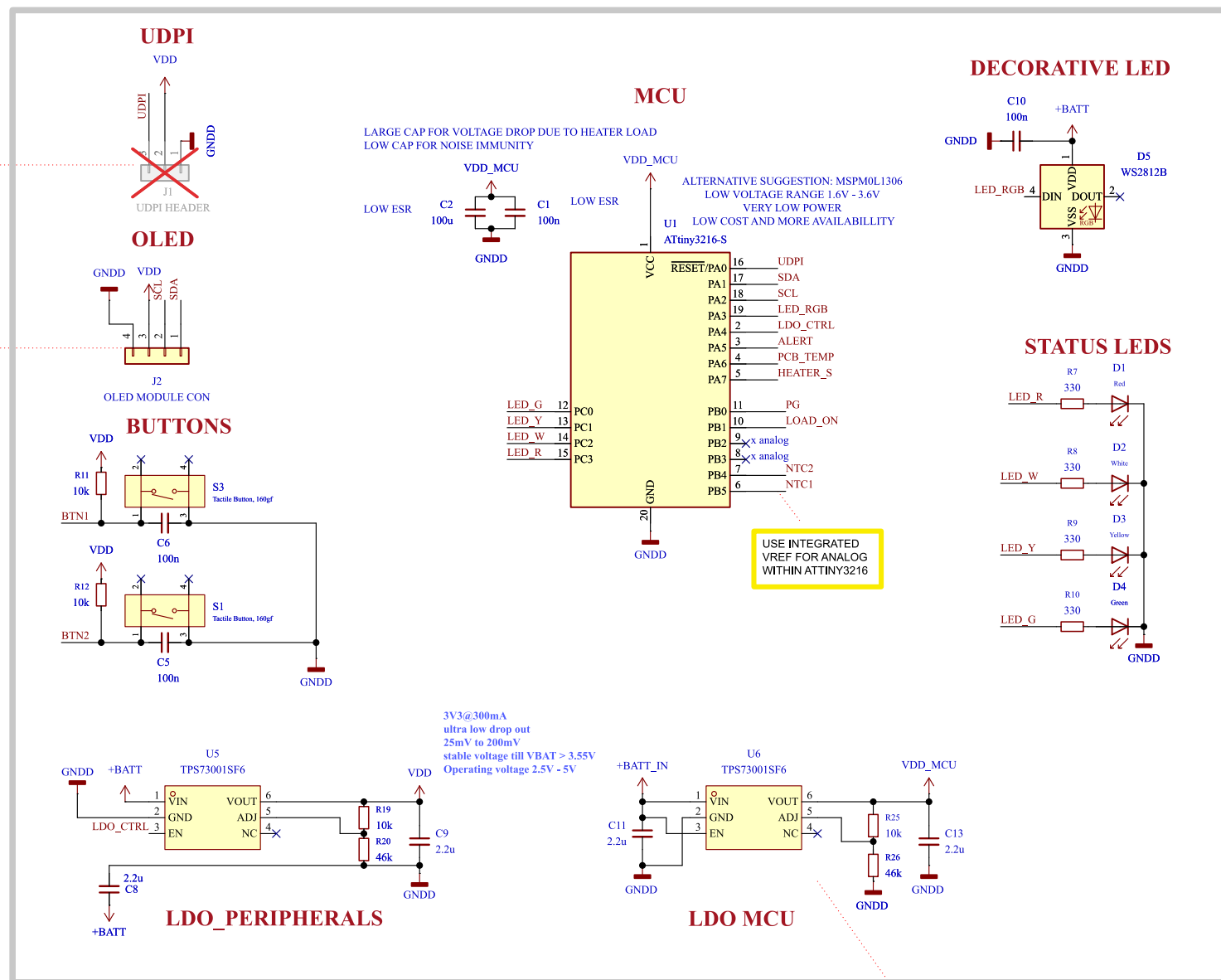
DESIGN NOTE: VDD will shift when voltage is below 3.3V causing inaccurate readings. below VDD1 firmware clauses have been added to handle this situation

According to section 18.2 of attiny3216 documentation internal VREF can be select if configured in the mcu which are more stable. It is recommended to use that as opposed to VDD domain

[illegible]

PWR GROUND TO
BE SEPERATED
WHILE ROUTING

[4]MAIN



LAYOUT NOTE:
INSTALL AT THE EDGE
FOR EASY ACCESS FOR
POGO PINS AND HEADER
PINS

**LAYOUT NOTE:
CREATE FOOTPRINT AND
ALLOW ADEQUATE
SPACE FOR INSTALLING
THE OLED MODULE**

USE INTEGRATED
VREF FOR ANALOG
WITHIN ATTINY3216

DESIGN NOTE:
VDD_MUC is always on
VDD_I_{DO} can be controlled by mcu

Table 5-1. PORT Function Multiplexing

Pin Name	Pin Name (1-3)	General/Special	DCDC1	ADCC1	PT0CN	AC5	AC1	AC2	DACS	USART0	SP0	TW0S	TCAC	TC0Bn	TC0D	CCL
23	PA0															
24	PA1	RESET/UPDI	AN0													LUT0-IN0
1	PA2		AN1													LUT0-IN1
18	PA3	EVOUT0	AN2													LUT0-IN2
2	PA4	EXTCLK	AN3													
3	PA5															
4	PA6															
5	PA7															
6	PA8															
7	PA9															
8	PA10															
9	PA11															
10	PA12															
11	PA13															
12	PA14															
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42	PA44															
43	PA45															
44	PA46															
45	PA47															
46	PA48															



OLED MODULE DATASHEET
Application note for INA234
J. Pieper ADC investigation

CAPISTOR TECHNOLOGIES - FZCO

HEATER CONTROLLER

Sheet Title A

Sheet Path:

Project Name:	HEATER CONTROLLER
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Designer:	Date:	Revision:
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SHOAIB MUSTAFA	Last Modified Date	1
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Reviewer:	Size:	Sheet:
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	1	2	3	4	5	6	7	8
A	[5] Sheet Title B							
B								
C								
D								
E								
F	1	2	3	4	5	6	7	8

	Comments:		Company:		Variant:		
	References:		CAPISTOR TECHNOLOGIES - FZCO		FIRST PROTOTYPE		
	Flexible I/O worked examples Flexible I/O source configuration		Board Name:		Project Name:		
			HEATER CONTROLLER		HEATER CONTROLLER		
Sheet Title:		File Name:		Designer:		Date:	Revision:
Sheet Title B		Section B - Ttle B.kicad_sch		SHOAIB MUSTAFA		Last Modified Date	1
Sheet Path:				Reviewer:		Size:	Sheet:
/Project Architecture/Section B - Title B/				SHOAIB MUSTAFA		A3	5 of 7

7 Parameter Measurement Information

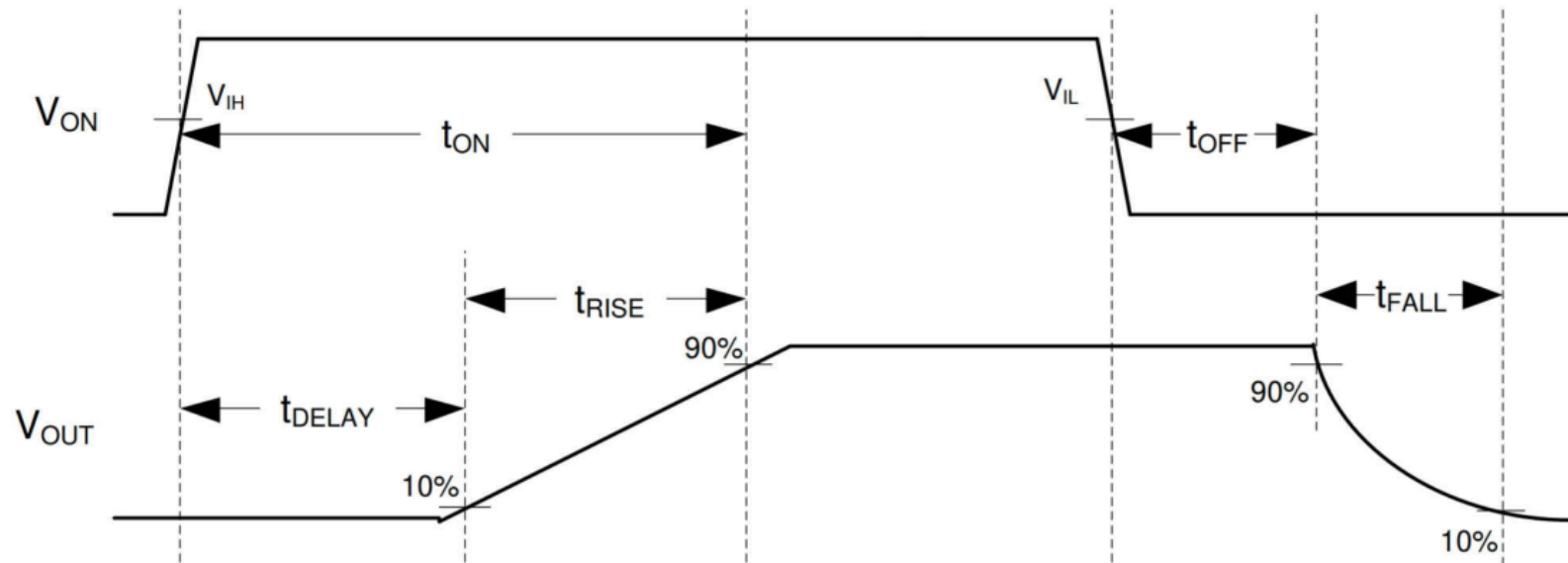


Figure 7-1. TPS22997 Timing Diagram

	Comments:	Company: CAPISTOR TECHNOLOGIES - FZCO		Variant: FIRST PROTOTYPE	
	REASONING FOR CHOOSING THIS METHOD	Board Name: HEATER CONTROLLER		Project Name: HEATER CONTROLLER	
	Sheet Title: Power - Sequencing	File Name: Power - Sequencing.kicad_sch	Designer: SHOAIB MUSTAFA	Date: Last Modified Date	Revision: 1
	Sheet Path: /Power - Sequencing/	Reviewer: SHOAIB MUSTAFA		Size: A4	Sheet: 6 of 7

[7] Revision History

DD.MM.YYYY - xxx Revision
Variant: xxx

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 capistor	Comments:		Company: CAPISTOR TECHNOLOGIES - FZCO		Variant: FIRST PROTOTYPE	
			Board Name: HEATER CONTROLLER		Project Name: HEATER CONTROLLER	
	Sheet Title: Revision History	File Name: Revision History.kicad_sch	Designer: SHOAIB MUSTAFA	Date: Last Modified Date	Revision: 1	
	Sheet Path: /Revision History/		Reviewer: SHOAIB MUSTAFA	Size: A4	Sheet: 7 of 7	